

Implemente todos los modelos estadísticos y de machine learning para predecir la variable de salida o respuesta: gastos Arbol de regresión

Muestra de entrenamiento y de prueba

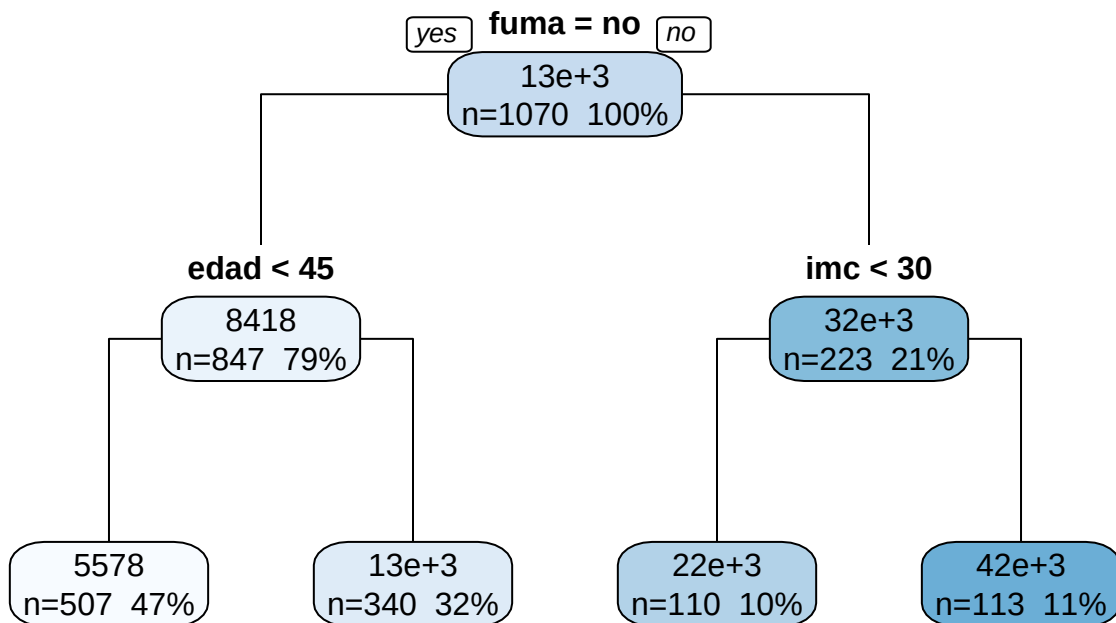
Modelo Arbol de desición

```
## Call:
## rpart(formula = gastos ~ ., data = s_train)
##   n= 1070
##
##           CP nsplit rel error   xerror   xstd
## 1 0.61320522    0 1.0000000 1.0031363 0.05756805
## 2 0.14407870    1 0.3867948 0.3895021 0.02122880
## 3 0.06505047    2 0.2427161 0.2471829 0.01629638
## 4 0.01000000    3 0.1776656 0.1876079 0.01502108
##
## Variable importance
##   fuma   imc  edad region  sexo  ninos
##    69    17    9     3     2     1
##
## Node number 1: 1070 observations,    complexity param=0.6132052
## mean=13277.69, MSE=1.462988e+08
## left son=2 (847 obs) right son=3 (223 obs)
## Primary splits:
##   fuma splits as LR,      improve=0.613205200, (0 missing)
##   edad < 41.5 to the left, improve=0.083284880, (0 missing)
##   imc < 30.6375 to the left, improve=0.043249470, (0 missing)
##   ninos < 1.5 to the left, improve=0.009901704, (0 missing)
##   region splits as LLRL,   improve=0.006630034, (0 missing)
##
## Node number 2: 847 observations,    complexity param=0.06505047
## mean=8417.71, MSE=3.691084e+07
## left son=4 (507 obs) right son=5 (340 obs)
## Primary splits:
##   edad < 44.5 to the left, improve=0.325715000, (0 missing)
##   ninos < 1.5 to the left, improve=0.026795680, (0 missing)
##   imc < 24.1775 to the left, improve=0.012563510, (0 missing)
##   region splits as RRLL,   improve=0.003858551, (0 missing)
##   sexo splits as RL,      improve=0.001213193, (0 missing)
## Surrogate splits:
##   imc < 35.885 to the left, agree=0.617, adj=0.047, (0 split)
##
## Node number 3: 223 observations,    complexity param=0.1440787
## mean=31736.88, MSE=1.313242e+08
## left son=6 (110 obs) right son=7 (113 obs)
## Primary splits:
##   imc < 29.9725 to the left, improve=0.770149400, (0 missing)
##   edad < 53.5 to the left, improve=0.137872300, (0 missing)
##   region splits as LLRL,   improve=0.021876010, (0 missing)
##   sexo splits as LR,      improve=0.008247271, (0 missing)
##   ninos < 1.5 to the left, improve=0.004654010, (0 missing)
## Surrogate splits:
##   region splits as LLRR,   agree=0.587, adj=0.164, (0 split)
##   sexo splits as LR,      agree=0.561, adj=0.109, (0 split)
```

```

##      edad  < 53.5    to the left,  agree=0.556, adj=0.100, (0 split)
##      ninos  < 0.5    to the right, agree=0.529, adj=0.045, (0 split)
##
## Node number 4: 507 observations
##   mean=5578.279, MSE=2.33223e+07
##
## Node number 5: 340 observations
##   mean=12651.8, MSE=2.722378e+07
##
## Node number 6: 110 observations
##   mean=21543.86, MSE=2.652424e+07
##
## Node number 7: 113 observations
##   mean=41659.28, MSE=3.374844e+07

```



Reglas de decision

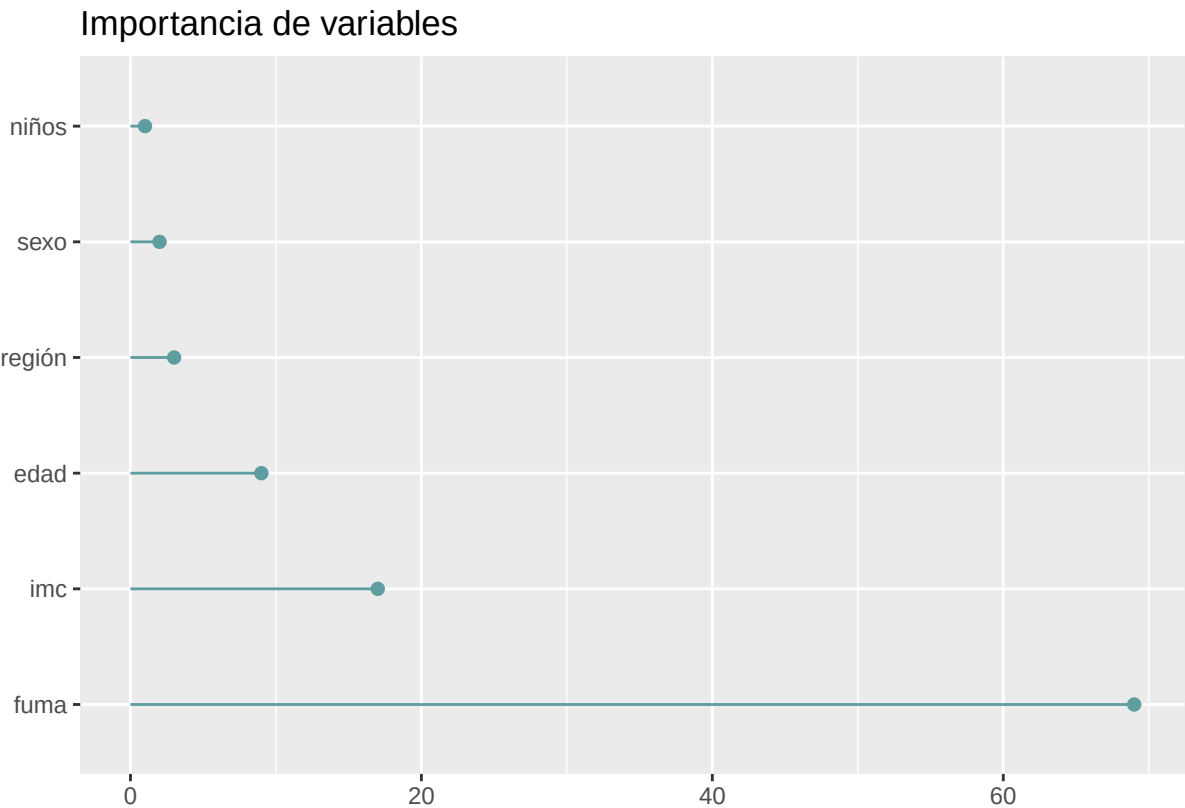
| gastos |      |      |    |     |   |      |    |    |             |
|--------|------|------|----|-----|---|------|----|----|-------------|
| 5578   | when | fuma | is | no  | & | edad | <  | 43 |             |
| 12651  | when | fuma | is | no  | & | edad | >= | 43 |             |
| 21543  | when | fuma | is | yes |   |      |    |    | & imc < 30  |
| 41659  | when | fuma | is | yes |   |      |    |    | & imc >= 30 |

Uso de la muestra de prueba, valor obtenido raiz del error cuadratico medio (rmse), debido a que la variable gastos es una variable continua y el modelo es un arbol de regresion

```
## [1] 4963.311
```

Importancia de las variables de regresion

```
## Warning: package 'ggplot2' was built under R version 4.2.3
```



Ajuste de hiper parametros o turning (funcion control)

```
## Call:
## rpart(formula = gastos ~ ., data = s_train, control = control)
##   n= 1070
##
##           CP nsplit rel error   xerror   xstd
## 1 0.613205218    0 1.0000000 1.0023755 0.05746965
## 2 0.144078700    1 0.3867948 0.3885684 0.02111167
## 3 0.065050469    2 0.2427161 0.3460596 0.02034647
## 4 0.007727090    3 0.1776656 0.2853302 0.01946226
## 5 0.006495817    4 0.1699385 0.2831081 0.01994439
## 6 0.002577239    5 0.1634427 0.2759286 0.01976095
## 7 0.001272738    6 0.1608655 0.2755063 0.01977507
## 8 0.001114087    7 0.1595927 0.2755412 0.01978523
## 9 0.001000000    8 0.1584786 0.2747517 0.01975842
##
## Variable importance
##   fuma   imc   edad region   sexo  niños
##    68    16    11     3     2     1
##
## Node number 1: 1070 observations,      complexity param=0.6132052
```

```

## mean=13277.69, MSE=1.462988e+08
## left son=2 (847 obs) right son=3 (223 obs)
## Primary splits:
##      fuma splits as LR,          improve=0.613205200, (0 missing)
##      edad < 41.5 to the left, improve=0.083284880, (0 missing)
##      imc < 30.6375 to the left, improve=0.043249470, (0 missing)
##      ninos < 1.5 to the left, improve=0.009901704, (0 missing)
##      region splits as LLRL,      improve=0.006630034, (0 missing)
##
## Node number 2: 847 observations, complexity param=0.06505047
## mean=8417.71, MSE=3.691084e+07
## left son=4 (507 obs) right son=5 (340 obs)
## Primary splits:
##      edad < 44.5 to the left, improve=0.325715000, (0 missing)
##      ninos < 1.5 to the left, improve=0.026795680, (0 missing)
##      imc < 24.1775 to the left, improve=0.012563510, (0 missing)
##      region splits as RRL,      improve=0.003858551, (0 missing)
##      sexo splits as RL,         improve=0.001213193, (0 missing)
## Surrogate splits:
##      imc < 35.885 to the left, agree=0.617, adj=0.047, (0 split)
##
## Node number 3: 223 observations, complexity param=0.1440787
## mean=31736.88, MSE=1.313242e+08
## left son=6 (110 obs) right son=7 (113 obs)
## Primary splits:
##      imc < 29.9725 to the left, improve=0.770149400, (0 missing)
##      edad < 40.5 to the left, improve=0.106452600, (0 missing)
##      region splits as LLRR,      improve=0.017554030, (0 missing)
##      sexo splits as LR,          improve=0.008247271, (0 missing)
## Surrogate splits:
##      region splits as LLRR,      agree=0.587, adj=0.164, (0 split)
##      sexo splits as LR,          agree=0.561, adj=0.109, (0 split)
##      edad < 53.5 to the left, agree=0.556, adj=0.100, (0 split)
##      ninos < 0.5 to the right, agree=0.529, adj=0.045, (0 split)
##
## Node number 4: 507 observations, complexity param=0.00772709
## mean=5578.279, MSE=2.33223e+07
## left son=8 (142 obs) right son=9 (365 obs)
## Primary splits:
##      edad < 22.5 to the left, improve=0.1022966000, (0 missing)
##      ninos < 0.5 to the left, improve=0.0998281300, (0 missing)
##      region splits as RRL,      improve=0.0126856600, (0 missing)
##      imc < 32.16 to the left, improve=0.0042582150, (0 missing)
##      sexo splits as RL,         improve=0.0009162619, (0 missing)
##
## Node number 5: 340 observations, complexity param=0.006495817
## mean=12651.8, MSE=2.722378e+07
## left son=10 (126 obs) right son=11 (214 obs)
## Primary splits:
##      edad < 51.5 to the left, improve=0.109857800, (0 missing)
##      region splits as LLRL,      improve=0.011055190, (0 missing)
##      ninos < 1.5 to the left, improve=0.009180818, (0 missing)
##      imc < 31.295 to the left, improve=0.003471524, (0 missing)
##      sexo splits as RL,          improve=0.002176391, (0 missing)

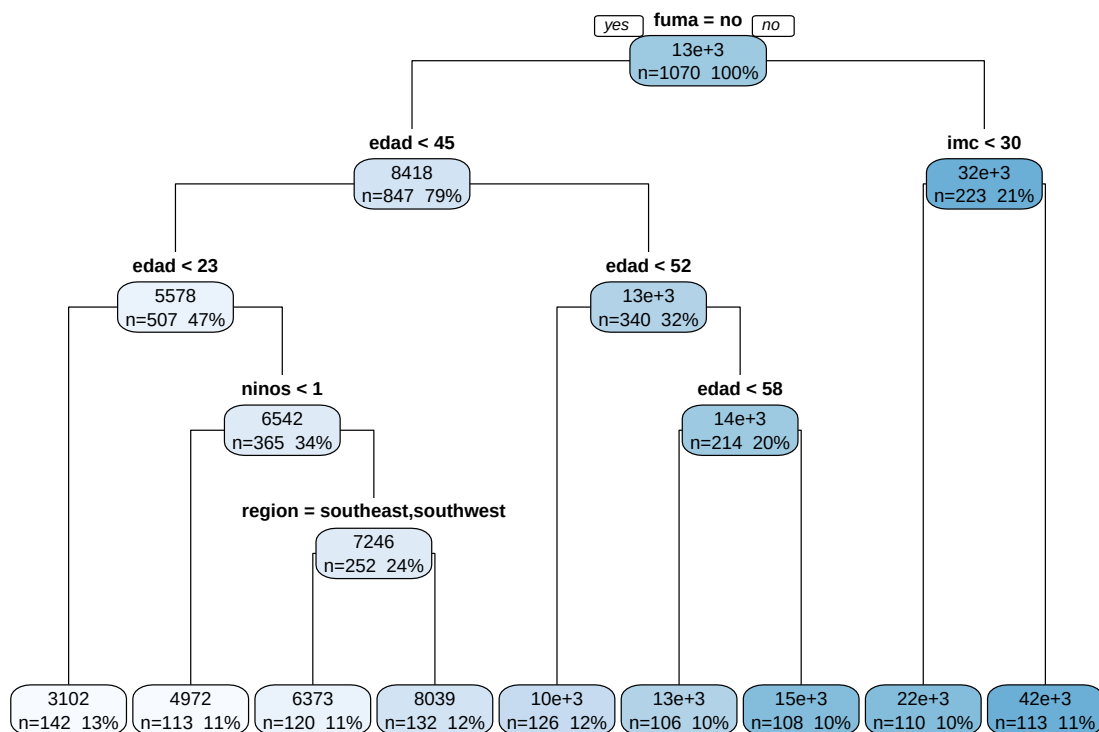
```

```

## Surrogate splits:
##      ninos < 0.5      to the right, agree=0.647, adj=0.048, (0 split)
##      imc  < 20.805   to the left,  agree=0.635, adj=0.016, (0 split)
##
## Node number 6: 110 observations
##   mean=21543.86, MSE=2.652424e+07
##
## Node number 7: 113 observations
##   mean=41659.28, MSE=3.374844e+07
##
## Node number 8: 142 observations
##   mean=3101.891, MSE=1.859508e+07
##
## Node number 9: 365 observations,      complexity param=0.002577239
##   mean=6541.695, MSE=2.184742e+07
##   left son=18 (113 obs) right son=19 (252 obs)
##   Primary splits:
##       ninos < 0.5      to the left,  improve=0.0505925200, (0 missing)
##       edad  < 32.5     to the left,  improve=0.0428616300, (0 missing)
##       region splits as  RRL,         improve=0.0189764000, (0 missing)
##       imc   < 33.29    to the left,  improve=0.0020813400, (0 missing)
##       sexo  splits as  RL,          improve=0.0003696295, (0 missing)
##   Surrogate splits:
##       edad < 24.5     to the left,  agree=0.721, adj=0.097, (0 split)
##
## Node number 10: 126 observations
##   mean=10398.02, MSE=1.984418e+07
##
## Node number 11: 214 observations,      complexity param=0.001272738
##   mean=13978.79, MSE=2.681713e+07
##   left son=22 (106 obs) right son=23 (108 obs)
##   Primary splits:
##       edad < 57.5     to the left,  improve=0.034716620, (0 missing)
##       region splits as  LRRL,        improve=0.024247750, (0 missing)
##       imc   < 31.295  to the left,  improve=0.009780750, (0 missing)
##       sexo  splits as  RL,          improve=0.002627346, (0 missing)
##   Surrogate splits:
##       ninos < 0.5      to the right, agree=0.607, adj=0.208, (0 split)
##       imc   < 34.3425  to the left,  agree=0.537, adj=0.066, (0 split)
##       region splits as  LRRL,        agree=0.537, adj=0.066, (0 split)
##       sexo  splits as  RL,          agree=0.519, adj=0.028, (0 split)
##
## Node number 18: 113 observations
##   mean=4971.678, MSE=1.832357e+07
##
## Node number 19: 252 observations,      complexity param=0.001114087
##   mean=7245.711, MSE=2.18266e+07
##   left son=38 (120 obs) right son=39 (132 obs)
##   Primary splits:
##       region splits as  RRL,         improve=0.031707130, (0 missing)
##       ninos < 1.5      to the left,  improve=0.021200500, (0 missing)
##       edad  < 33.5     to the left,  improve=0.020014570, (0 missing)
##       sexo  splits as  RL,          improve=0.003084388, (0 missing)
##       imc   < 29.595   to the left,  improve=0.002834703, (0 missing)

```

```
## Surrogate splits:
##   imc < 28.905 to the right, agree=0.631, adj=0.225, (0 split)
##   sexo splits as LR, agree=0.563, adj=0.083, (0 split)
##   ninos < 1.5 to the left, agree=0.563, adj=0.083, (0 split)
##   edad < 42.5 to the right, agree=0.552, adj=0.058, (0 split)
##
## Node number 22: 106 observations
##   mean=13004.85, MSE=2.686394e+07
##
## Node number 23: 108 observations
##   mean=14934.7, MSE=2.492643e+07
##
## Node number 38: 120 observations
##   mean=6373.206, MSE=1.218824e+07
##
## Node number 39: 132 observations
##   mean=8038.897, MSE=2.926755e+07
```



RMSE

```
pred1 <- predict(object = s_model1, newdata = s_test)
sqrt(mean((pred1 - s_test$gastos)^2))
```

```
## [1] 4892.29
```

| gastos | V2   | V3   | V4 | V5  | V6 | V7   | V8 | V9 | V10 | V11 | V12 | V13   | V14 | V15 | V16 | V17 | V18 |
|--------|------|------|----|-----|----|------|----|----|-----|-----|-----|-------|-----|-----|-----|-----|-----|
| 3102   | when | fuma | is | no  | &  | edad | <  | 23 |     |     |     |       |     |     |     |     |     |
| 4972   | when | fuma | is | no  | &  | edad | is | 23 | to  | 45  | &   | nicos | <   | 1   |     |     |     |
| 6373   | when | fuma | is | no  | &  | edad | is | 23 | to  | 45  | &   | nicos | >=  | 1   |     |     |     |
| 8039   | when | fuma | is | no  | &  | edad | is | 23 | to  | 45  | &   | nicos | >=  | 1   |     |     |     |
| 10398  | when | fuma | is | no  | &  | edad | is | 45 | to  | 52  |     |       |     |     |     |     |     |
| 13005  | when | fuma | is | no  | &  | edad | is | 52 | to  | 58  |     |       |     |     |     |     |     |
| 14935  | when | fuma | is | no  | &  | edad | >= |    |     | 58  |     |       |     |     |     |     |     |
| 21544  | when | fuma | is | yes |    |      |    |    |     |     |     |       |     |     | &   | imc | <   |
| 41659  | when | fuma | is | yes |    |      |    |    |     |     |     |       |     |     | &   | imc | >=  |